

weeks-1-and-2\1st-math-problems.py

```
1
2 '''
3 Algebra week two
4 '''
5
6 '''
7 Example problem notation:
8 '''
9
10  $3(x - 2) + 4 = 19$ 
11
12 #step one: distribute the 3 to the (x-2)
13  $3x - 6 + 4 = 19$ 
14
15 #step two: combine constants
16  $-6 + 4 = -2$ 
17  $3x - 2 = 19$ 
18
19 #step three: add 2 to both sides
20  $(3x - 2) + 2 = (19) + 2$ 
21  $-2 + 2 = 0$ ;  $19 + 2 = 21$ 
22  $(3x) = (21)$ 
23
24 #step four: divide both sides by 3
25  $(3x) / 3 = (21) / 3$ 
26  $\#3 / 3 = 1$ ;  $21 / 3 = 7$ 
27  $x = 7$ 
28
29 '''
30 problem 1
31 '''
32
33  $f(x) = -3x^3 - 2$ 
34  $g(x) = 3x + 2$ 
35
36 Find  $f(-4)$  and  $g(-3)$ 
37
38 #step one: substitute -4 for x in f(x)
39  $f(-4) = -3(-4)^3 - 2$ 
40  $= -3(-64) - 2$ 
41  $= 192 - 2$ 
42  $= 190$ 
43
44 #step two: substitute -3 for x in g(x)
45  $g(-3) = 3(-3) + 2$ 
46  $= -9 + 2$ 
47  $= -7$ 
48
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49 #step three: final answers
50  $f(-4) = 190$ 
51  $g(-3) = -7$ 
52
53 '''
54 Solving a linear equation with several occurrences of the variable: Variables on both sides
55 and distribution
56 '''
57
58  $7x - 15 = 3(x - 1)$ 
59
60 #step one: distribute the 3 to the (x-1)
61  $7x - 15 = 3x - 3$ 
62
63 #step two: subtract 3x from both sides
64  $(7x - 15) - 3x = (3x - 3) - 3x$ 
65  $\#7x - 3x = 4x$ ;  $3x - 3x = 0$ 
66  $4x - 15 = -3$ 
67
68 #step three: add 15 to both sides
69  $(4x - 15) + 15 = (-3) + 15$ 
70  $\#-15 + 15 = 0$ ;  $-3 + 15 = 12$ 
71  $4x = 12$ 
72
73 #step four: divide both sides by 4
74  $(4x / 4) = (12 / 4)$ 
75  $\#4 / 4 = 1$ ;  $12 / 4 = 3$ 
76  $1x = 3$ 
77
78 '''
79 Graph the following functions
80 '''
81
82  $f(x) = x$ 
83  $f(x) = (x^2)$ 
84  $f(x) = |x|$ 
85  $f(x) = (x^3)$ 
86
87  $f(x) = x$ 
88 #step one: create a table of values for both: x and f(x)
89
90 x | f(x)
91 -2 | -2
92 -1 | -1
93 0 | 0
94 1 | 1
95 2 | 2
96
97 '''
98 graph is done in desmos; it's a straight line through the (0, 0) origin with a slope of rise:
99 1 / run: 1
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97 '''
98 f(x) = (x^2)
99 #step one: create a table of values for both: x and f(x)
100 #f(x) = each x value is squared
101 x | f(x)
102 -2 | 4
103 -1 | 1
104 0 | 0
105 1 | 1
106 2 | 4
107
108 '''
109 graph is done in desmos; it's a skinny parabola opening **upwards** with the vertex (center)
    at the (0, 0) origin
110 why is the parabola skinny? because the coefficient of x^2 is 1, which is a small number; if
    it were a larger number, the parabola would be wider
111 so if the graph was f(x) = 10x^2, the parabola would be fatter than f(x) = x^2
112 '''
113
114 f(x) = |x|
115
116 #step one: create a table of values for both: x and f(x)
117 x | f(x)
118 -2 | 2
119 -1 | 1
120 0 | 0
121 1 | 1
122 2 | 2
123
124 '''
125 graph is done in desmos; it's a v-shaped graph with the vertex (center) at the (0, 0) origin
126
127 why is it v-shaped? because the absolute value of a number is always positive, so the graph
    will always be above the x-axis
128
129 so the line will be on the left if the y-axis is negative, and on the right if the y-axis is
    positive
130 but it will never be on the bottom since the x-axis will always be positive
131 '''
132
133 '''
134 Calculate the distance between the points in the coordinate plane
135 '''
136
137 E = (0,7) and M = (6,-1)
138
139 #step one: use the distance formula to find the distance between the two points
140
141 distance =  $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ 
142
143 distance =  $\sqrt{(6 - 0)^2 + (-1 - 7)^2}$ 

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144 =  $\sqrt{6^2 + -8^2}$ 
145 =  $\sqrt{36 + 64}$ 
146 =  $\sqrt{100}$ 
147 = 10
148
149 '''
150 Calculate the distance between the two points
151 '''
152
153 G = (5,-9) and F = (8,-1)
154
155 #step one: use the distance formula
156
157 distance =  $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ 
158
159 distance =  $\sqrt{(8 - 5)^2 + (-1 - (-9))^2}$ 
160 =  $\sqrt{3^2 + (-1 + 9)^2}$ 
161 =  $\sqrt{9 + (8)^2}$ 
162 =  $\sqrt{9 + 64}$ 
163 =  $\sqrt{73}$ 
164
165 '''
166 Calculate the distance between the two points
167 '''
168
169 L = (-7,-8) and H = (1,-1)
170
171 #step one: use the distance formula
172
173 distance =  $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ 
174
175 distance =  $\sqrt{(1 - (-7))^2 + (-1 - (-8))^2}$ 
176 =  $\sqrt{(1 + 7)^2 + (-1 + 8)^2}$ 
177 =  $\sqrt{64 + 49}$ 
178 =  $\sqrt{113}$ 
179
180 '''
181 Evaluate
182 '''
183 -(-4)3 and -(7)2
184
185
186 -(-4)3
187 -(-64)
188 64
189
190 -(7)2
191 -(49)
192
193

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194 '''
195 Solve for v; simplify your answer
196 '''
197 -4(v + 3) = 4v + 28
198
199
200 #step one: distribute -4 on left
201 (-4v) + (-12) = 4v + 28
202
203
204 #step two: add 4v to both sides
205 (4v) + (-4v) + (-12) = (4v) + 4v + 28
206
207
208 #step three: combine like terms
209 0 + (-12) = 8v + 28
210
211
212 #step four: divide both sides by 8
213 (-12) / 8 = ((8v) / 8) + (28 / 8)
214
215
216 #step five: reduce terms
217 (-3)/2 = 1v + (7/2)
218
219
220 #step six: subtract (7/2) from both sides
221 ((-3)/2) + ((-7)/2) = (1v + (7/2)) + ((-7)/2)
222
223 ((-10)/2) = 1v
224
225
226 #reduce
227 (-5)/1 = 1v
228 (-5) = v
229
230
231 '''
232 simplify
233 '''
234 ((v)5 (u)7) / ((v)4 (u)1)
235
236 #v's expnts: 5 - 4 = 1
237 #u's expnts: 7 - 1 = 6
238 #this expnt subtraction removes the denominator
239
240 (v)1 (u)6
241 v (u)6
242
243 '''

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244 simplify
245 '''
246 ((x)6 (z)4) / (x)6 (z)5
247
248 #x's expnts: 6 - 6 = 0
249 #z's expnts: 4 - 5 = (-1)
250
251 (x)0 (z)-1
252
253 #(x)0 = 1
254 #(z)-1 = (1/z)
255
256 1 (1 / z)
257 1/z
258
259 '''
260 simplify
261 '''
262
263 ((a)4 (b)2) / ((a)3 (b)6)
264
265 #(a) 4 - 3 = 1
266 #(b) 2 - 6 = (-4)
267
268 (a)1 (b)-4
269
270 #(a) = (a) / 1
271 #(b)-4 = (1/b4)
272 ((a) / 1) (1 / b4)
273
274 #neither the 1 in the denominator of (a), nor the 1 in the numerator of (b4) is necessary. so
we remove them and then we're left with (a) in the numerator and (b4) in the denominator
275 (a) / b4
276
277 '''
278 simplify
279 '''
280 ((b)6 (a)3) / ((b)7 (a))
281
282 #(b) 6 - 7 = -1
283 #(a) 3 - 1 = 2
284
285 (b)-1 (a)2
286
287 #(b)-1 = (1/b1) = (1/b)
288 #see note in line 274
289
290 (a)2 / (b)
291
292 '''

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293 simplify
294 '''
295  $((x)^4 (z)^5) / (3(x)^4 (z)^3)$ 
296
297  $\#(x)^4 - (x)^4 = (x)^0 = (1)$ 
298  $\#(z)^5 - (z)^3 = (z)^2$ 
299
300  $((1) (z)^2) / 3$ 
301  $(z)^2 / 3$ 
302
303 '''
304 simplify
305 '''
306
307  $((a)^7 (b)^6) / ((a) (b)^4)$ 
308
309  $\#(a)^7 - (a)^1 = (a)^6$ 
310  $\#(b)^6 - (b)^4 = (b)^2$ 
311
312  $(a)^6 (b)^2$ 
313
314 '''
315 simplify
316 '''
317
318  $((b)^7 (a)) / ((b)^6 (a)^4)$ 
319
320  $\#(b)^7 - (b)^6 = (b)^1$ 
321  $\#(a)^1 - (a)^4 = (a)^{-3}$ 
322  $(b)^1 (a)^{-3}$ 
323
324  $\#(a)^{-3} = (1 / a^3)$ 
325  $(b) / (a^3)$ 
326
327 '''
328 find the value
329 '''
330  $\sqrt[3]{8}$ 
331
332 #what number = 8 after being cubed?
333  $\#(2)^3 = 8$ 
334 2
335
336 '''
337 find the value
338 '''
339  $\sqrt[3]{-216}$ 
340
341 #what number = -216 after being cubed?
342 #because the answer is negative, I need to cube a negative number

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343  $\#(-6)^3 = -216$ 
344 -6
345
346 '''
347 rewrite w/out parentheses and simplify
348 '''
349  $(x + 2)^2$ 
350
351 #shortcut formula --->
352 #square the first term + double the product of the two terms + square the second term
353
354  $\#(a + b)^2 = a^2 + 2ab + b^2$ 
355 #a = x
356 #b = 2
357
358  $(x)^2 + 2(x)(2) + (2)^2$ 
359  $(x)^2 + 4(x) + 4$ 
360
361 '''
362 rewrite w/out parentheses and simplify
363 '''
364  $(4u - 5)^2$ 
365
366 #shortcut formula --->
367 #square the first term - double the product of the two terms + square the second term
368
369  $\#(a - b)^2 = a^2 - 2ab + b^2$ 
370 #a = 4u
371 #b = 5
372
373  $(4u)^2 - 2(4u)(5) + (5)^2$ 
374  $(16u^2) - 2(20u) + 25$ 
375  $(16u^2) - 40u + 25$ 
376
377 '''
378 rewrite w/out parentheses and simplify
379 '''
380
381  $(2 + 3y)^2$ 
382  $(3y + 2)^2$ 
383
384 #shortcut formula --->
385 #square the first term + double the product of the two terms + square the second term
386
387  $\#(a + b)^2 = a^2 + 2ab + b^2$ 
388 #a = 3y
389 #b = 2
390
391  $(3y)^2 + 2(3y)(2) + (2)^2$ 
392  $(9y)^2 + 2(6y) + 4$ 

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393 (9y)2 + (12y) + 4
394
395 '''
396 simplify the square root of a whole number < 100
397 '''
398
399  $\sqrt{8}$ 
400
401 #find the largest perfect square factor ---> 4
402 8 = 4 * 2
403
404 #rewrite square root of 8 as square root of 4 * 2
405  $\sqrt{8} = \sqrt{4 \times 2}$ 
406
407 #separate the square roots
408  $\sqrt{8} = \sqrt{4} * \sqrt{2}$ 
409
410 #simplify  $\sqrt{4}$ 
411  $\sqrt{8} = 2 * \sqrt{2}$ 
412 2 *  $\sqrt{2}$ 
413
414 '''
415 simplify the square root of a whole number < 100
416 '''
417  $\sqrt{27}$ 
418
419 #find the largest perfect square factor ---> 9
420 27 = 9 * 3
421
422 #rewrite square root of 27 as square root of (9 * 3)
423
424  $\sqrt{27} = \sqrt{9 * 3}$ 
425
426 #separate the square roots
427  $\sqrt{27} = \sqrt{9} * \sqrt{3}$ 
428
429 #simplify  $\sqrt{9}$ 
430  $\sqrt{27} = 3 * \sqrt{3}$ 
431 3 *  $\sqrt{3}$ 
432
433
434 '''
435 simplify the square root of a whole number < 100
436 '''
437  $\sqrt{32}$ 
438
439 #find the largest perfect square factor ---> 16
440 32 = 16 * 2
441
442 #rewrite square root of 32 as square root of (16 * 2)

```

```

443
444  $\sqrt{32} = \sqrt{16 * 2}$ 
445
446 #separate the square roots
447  $\sqrt{32} = \sqrt{16} * \sqrt{2}$ 
448
449 #simplify  $\sqrt{16}$ 
450  $\sqrt{32} = 4 * \sqrt{2}$ 
451 4 *  $\sqrt{2}$ 
452
453

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